

Ejercicios Sección 9.1

9.1)

$$a) \theta = \frac{s}{r} = \frac{1.5 \text{ m}}{2.5 \text{ m}} = \underline{0.6 \text{ rad}}$$

$$0.6 \text{ rad} \times \frac{180^\circ}{\pi \text{ rad}} = \underline{34.38^\circ}$$

$$b) 128^\circ \times \frac{\pi}{180^\circ} = 2.23 \text{ rad}$$

$$2.23 \text{ rad} = \frac{14 \text{ cm}}{r} \quad r = \frac{14 \text{ cm}}{2.23 \text{ rad}} = \underline{6.27 \text{ cm}}$$

$$c) 0.700 \text{ rad} = \frac{s}{1.5 \text{ m}} \quad s = 0.700 \text{ rad} \times 1.5 \text{ m} = \underline{1.05 \text{ m}}$$

$$9.3) \omega_z(t) = A + Bt^2 \quad \omega_z(t) = 2.75 + 1.5t^2$$

$$A = \text{rad/seg}$$

$$B = \text{rad/seg}^3$$

$$b) \alpha(t) = 2Bt = 3 \text{ rad/seg}^2 \quad i) \alpha_z(t) = 0$$

$$ii) t = 5 \text{ s} \quad 3 \frac{\text{rad}}{\text{seg}^2} = \frac{\omega}{5} \quad \omega = 3 \cdot 5 = \underline{15 \text{ rad/seg}^2}$$

$$c) \theta - \theta = (2.75)(2) + \frac{1}{3}(1.5)(2)^3 = 9.5 \text{ rad}$$

$$9.5) \theta(t) = \gamma t + \beta t^3 \quad \theta(t) = 0.400 \text{ rad/s} t + 0.0120 \text{ rad/s}^3$$

$$a) \omega = \frac{d\theta}{dt} = \gamma + 3\beta t^2$$

$$b) t=0 \quad \omega = \gamma = 0.400 \text{ rad/s}$$

$$c) t=5s \quad \omega_{med} = t=0 \text{ a } t=5$$

$$\omega = 0.400 + 3(0.0120)(5)^2 = 1.3 \text{ rad/s}$$

$$\theta = (0.400)(5) + (0.0120)(5)^3 = 3.5 \text{ rad}$$

$$\omega = \frac{3.5 - 0}{5 - 0} = 0.7 \text{ rad/s}$$

$$9.7) \theta(t) = a + bt - ct^3$$

$$t=0 \quad \theta = \frac{\pi}{4} \quad \omega = 2 \text{ rad/s}$$

$$t=1.5s \quad \alpha = 1.25 \text{ rad/s}^2$$

$$a) \omega(t) = b - 3ct^2 = b = 2 \text{ rad/s} \quad t=1.5s$$

$$\alpha(t) = -6ct = -6c(1.5) = 1.25 \text{ rad/s}^2 \quad c = -0.139 \text{ rad/s}^3$$

$$t=0 \quad \theta = a = \frac{\pi}{4} \text{ rad}$$

$$b) \theta = \frac{\pi}{4} \text{ rad} \quad \alpha = 0 \quad t=0$$

Norma

$$c) \alpha_z = 3.5 \text{ rad/s}^2 \quad t = -\frac{\omega_z}{\alpha_z} = -\frac{3.5 \text{ rad/s}}{6(-0.139 \text{ rad/s}^2)} = 4.20 \text{ s}$$

$$t = 4.2 \text{ s} \quad \theta = \frac{\pi}{4} + (2)(4.2) - (-0.139)(4.20)^3 = 19.48 \text{ rad}$$

$$\omega_z = 2 - 3(-0.139)(4.20)^2 = 9.36 \text{ rad/s}$$

$$9.9) \omega_0 = 1.5 \text{ rad/s}$$

$$a) \alpha = 0.300 \text{ rad/s}^2 \quad t = 2.5 \text{ s}$$

$$\omega_z = 1.5 \frac{\text{rad}}{\text{s}} + 0.300 \frac{\text{rad}}{\text{s}^2} (2.5 \text{ s}) = 2.25 \text{ rad/s}$$

$$b) \theta = ? \quad t = 0 \quad t = 2.5$$

$$\theta - \theta_0 = (1.5 \frac{\text{rad}}{\text{s}})(2.5 \text{ s}) + \frac{1}{2}(0.300)(2.5)^2 = 4.69 \text{ rad}$$

$$9.11) \alpha = 1.5 \text{ rad/s}^2 \quad \omega = 36 \text{ rad/s}$$

$$a) 36 \frac{\text{rad}}{\text{s}} = 0 + (1.5 \frac{\text{rad}}{\text{s}^2}) t$$

$$\frac{36}{1.5} = t = 24 \text{ s}$$

$$b) \theta - \theta_0 = 0 + \frac{1}{2}(1.5)(24)^2 = 432 \text{ rad}$$

$$\frac{432 \text{ rad}}{2\pi \text{ rad}} = 68.8 \text{ rev}$$

$$9.13) \alpha = 2.25 \text{ rad/s}^2 \quad t = 4 \text{ s} \quad \theta = 60 \text{ rad}$$

$$60 \text{ rad} = \omega_{0z}(4 \text{ s}) + \frac{1}{2}(2.25 \text{ rad/s}^2)(4)^2$$

$$60 = 4\omega_{0z} + 18$$

$$\frac{60 - 18}{4} = \omega_{0z} = \underline{10.5 \text{ rad/s}}$$

$$9.15) \omega_{0z} = 500 \text{ rpm} \quad m = 40 \text{ kg} \quad d = 75 - r = 37.5 \text{ cm}$$

$$t = 30 \text{ s} \quad \theta - \theta_0 = 200 \text{ rev}$$

$$500 \text{ rpm} = 8.333 \text{ rev/s}$$

$$\theta - \theta_0 = \frac{1}{2}(\omega_{0z} + \omega_z)t$$

$$200 \text{ rev} = \frac{1}{2}(8.333 + \omega_z)30$$

$$200 = \left(\frac{8.333 + \omega_z}{2} \right) 30$$

$$\omega_z = 5 \text{ rev/s} = \underline{300 \text{ rpm}}$$

$$b) \omega_z = \omega_{0z} + \alpha_z t$$

$$5 = 8.333 + \alpha_z(30)$$

$$\frac{5 - 8.333}{30} = \alpha_z = \underline{-0.1111 \text{ rev/s}^2}$$

$$\omega_z = \omega_{0z} + \alpha_z t$$

$$0 = 8.333 + (-0.1111)t$$

$$\frac{-8.333}{-0.1111} = t = \underline{75 \text{ s}}$$

$$0.1111$$

$$\theta - \theta_0 = \left(\frac{\omega_0 + \omega}{2} \right) t$$

$$\theta - \theta_0 = \left(\frac{8.333 + 0}{2} \right) (75)$$

$$\theta - \theta_0 = \underline{312.49 \text{ rev}}$$

$$a) \theta = 1 \text{ rev} = 2\pi \quad t = 3.156 \times 10^7 \text{ s}$$

$$\omega = \frac{2\pi}{3.156 \times 10^7} = 1.99 \times 10^{-7} \text{ rad/s}$$

$$b) \theta = 1 \text{ rev} = 2\pi \quad t = 86,400 \text{ s}$$

$$\omega = \frac{2\pi}{86,400} = 7.27 \times 10^{-5} \text{ rad/s}$$

$$c) v = r\omega \quad v = (1.5 \times 10^{11} \text{ m})(1.99 \times 10^{-7}) = 2.98 \times 10^4 \text{ m/s}$$

$$d) v = r\omega \quad v = (6.38 \times 10^6 \text{ m})(7.27 \times 10^{-5}) = 464 \text{ m/s}$$

$$e) a = r\omega^2 \quad v = (6.38 \times 10^6 \text{ m})(7.27 \times 10^{-5}) = 0.0337 \text{ m/s}^2$$

$$a = r\alpha = 0 \quad \alpha = 0$$

Norma

$$9.21) a) d = 40 \text{ cm} \quad \alpha = 3 \text{ rad/s}^2 \quad \text{rev} = 2$$

$$2 \text{ rev} = 4\pi \text{ rad} \quad r = 0.2 \text{ m} \quad a_{\text{rad}} = \omega^2 r$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

$$\omega_z = \sqrt{2(3 \text{ rad/s}^2)(4\pi)} = 8.68 \text{ rad/s} \quad (\text{Eq. 1})$$

$$a_{\text{rad}} = (8.68 \text{ rad/s})^2 (0.2 \text{ m}) = \underline{15 \text{ m/s}^2}$$

$$b) a_{\text{rad}} = \frac{v^2}{r}$$

$$v = r\omega$$

$$v = (0.2 \text{ m})(8.68 \text{ rad/s}) = 1.74 \text{ m/s}$$

$$a_{\text{rad}} = \frac{(1.74 \text{ m/s})^2}{0.2 \text{ m}} = \underline{15 \text{ m/s}^2}$$

$$9.23) r = 0.300 \text{ m} \quad \alpha = 0.600 \text{ m/s}^2$$

$$a) a_{\text{tan}} = r\alpha = (0.300 \text{ m})(0.600 \text{ m/s}^2) = 0.18 \text{ m/s}^2$$

$$a_{\text{rad}} = r\omega^2 = 0 \quad a = \sqrt{a_{\text{rad}}^2 + a_{\text{tan}}^2} = 0.18 \text{ m/s}^2$$

$$= 0 + 0.18$$

$$b) \theta - \theta_0 = 60^\circ \neq \frac{\pi}{180^\circ} = 1.047 \text{ rad}$$

$$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$$

$$\omega_z = \sqrt{2(0.600 \text{ m/s}^2)(1.047 \text{ rad})} = 1.12 \text{ rad/s}$$

9.25) $m = 3.000 \text{ kg}$ $\omega = 5000 \text{ rev/min}$

$$a_{\text{rad}} = r\omega^2 \quad r = \frac{a_{\text{rad}}}{\omega^2} \quad a = (3.000)(9.8 \text{ m/s}^2) = 29,400$$

$$5000 \frac{\text{rev}}{\text{min}} \times \frac{1 \text{ min}}{60 \text{ s}} \times \frac{2\pi \text{ rad}}{1 \text{ rev}} = 523.6 \text{ rad/s} = \omega$$

$$r = \frac{29,400}{523.6} = 0.107 \text{ m}$$

9.27) $d = 12.7 \text{ mm}$ $V = 1250 \text{ rev/min}$
 $V =$

$$V = r\omega$$

$$a) 1250 \frac{\text{rev}}{\text{min}} \times \frac{\pi \text{ rad}}{30 \text{ rev/min}} \times \frac{12.7 \times 10^{-3} \text{ m}}{2} = 0.831 \text{ m/s} = V$$

$$b) \frac{V^2}{r} = \frac{0.831^2}{(12.7 \times 10^{-3})/2} = 109 \text{ m/s}^2$$

9.29) a) $a_{\text{rad}} = r\omega^2$

$$F = ma = mr\omega^2$$

$$\frac{F_2}{F_1} = \left(\frac{\omega_2}{\omega_1} \right)^2 = \frac{640 \text{ rev/min}}{423 \text{ rev/min}} = 2.29$$

$$b) V = r\omega$$

$$\frac{V_2}{V_1} = \frac{\omega_2}{\omega_1} = \frac{640 \text{ rev/min}}{423 \text{ rev/min}} = 1.51$$

$$c) V = r\omega$$

$$\omega = 640 \frac{\text{rev}}{\text{min}} \cdot \frac{1 \text{ min}}{60 \text{ s}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} = 67 \text{ rad/s}$$

9.37) a) Varilla

i) Por un extremo, $I = \frac{1}{3} ML^2 = \frac{1}{3} (2.5 \text{ Kg})(0.75 \text{ m})^2 = 0.47 \text{ Kg/m}^2$
 $75 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.75$

ii) Por su centro, $I = \frac{1}{12} ML^2 = \frac{1}{12} (2.5)(0.75)^2 = 0.12 \text{ Kg/m}^2$

iii) $I = 0$ toda la masa está en el eje

b) Esfera

i) Sólida $I = \frac{2}{5} MR^2 = \frac{2}{5} (3 \text{ Kg})(0.19 \text{ m})^2 = 0.04 \text{ Kg/m}^2$
 $38 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.38/2 = 0.19 \text{ m}$

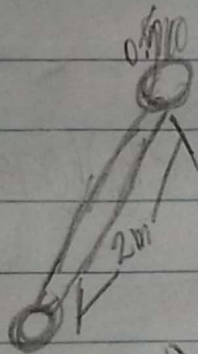
ii) Hueca y pared delgada $I = \frac{2}{3} MR^2 = \frac{2}{3} (3)(0.19)^2 = 0.07 \text{ Kg/m}^2$

c) Cilindros

i) Al usar ejes delgado $I = MR^2 = (8)(0.06)^2 = 0.03 \text{ Kg/m}^2$
 $6 \text{ cm} \cdot \frac{1 \text{ m}}{100 \text{ cm}} = 0.06$

ii) Sólido $I = \frac{1}{2} MR^2 = \frac{1}{2} (8)(0.06)^2 = 0.014 \text{ Kg/m}^2$

9.33) $L = 2 \text{ m}$ $M = 4 \text{ Kg}$
 $M = 0.500 \text{ Kg}$



a) $I = I_{\text{barra}} + I_{\text{esferas}} = \frac{1}{12} ML^2 + 2M\left(\frac{L}{2}\right)^2$
 $= \frac{1}{12} (4 \text{ Kg}) (2 \text{ m})^2 + 2(0.500 \text{ Kg}) (1 \text{ m})^2 = 2.33 \text{ Kg/m}^2$

b) $I = \frac{1}{3} M_{\text{esf}} L^2 + M_{\text{esf}} L^2 = \frac{1}{3} (4)(2)^2 + (0.500)(2)^2 = 7.33 \text{ Kg/m}^2$

c) Las masas están en el ej = $I = 0$

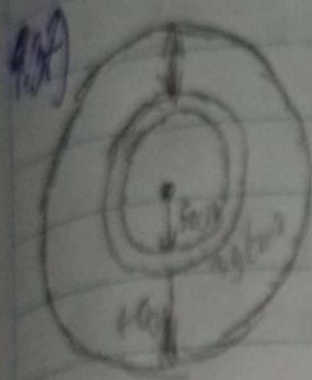
d) $I = M_{\text{esf}} d^2 + 2M_{\text{esf}} d^2 = M_{\text{esf}} d^2 = (5 \text{ Kg})(0.5 \text{ m})^2 = 1.25 \text{ Kg/m}^2$

9.35 $I = \sum mr^2$ $I = I_{\text{ex}} + I_{\text{rot}}$

$I_{\text{ex}} = MR^2 = (1.4 \text{ Kg})(0.300 \text{ m})^2 = 0.13 \text{ Kg/m}^2$

$I_{\text{rot}} = 8\left(\frac{1}{3} ML^2\right) = \frac{8}{3} (0.200 \text{ Kg})(0.300)^2 = 0.067$

$$I = I_{ext} + I_{rot} = 0.13 + 0.061 = \underline{0.19 \text{ Kg/m}^2}$$



$$I = M R^2$$

$$I = \frac{1}{2} M (r_1^2 + r_2^2)$$

$$\text{disco} = m = 3\pi r^2 = 23.56 \text{ Kg}$$

$$I = \frac{1}{2} M R^2 = \frac{1}{2} (23.56) (0.50 \text{ m})^2 = 2.95 \text{ Kg/m}^2$$

$$\text{anillo} = m = 2\pi (r_2^2 - r_1^2) = 15.08 \text{ Kg}$$

$$I = \frac{1}{2} M (r_1^2 + r_2^2) = 5.58 \text{ Kg/m}^2$$

$$I_t = 2.95 + 5.58 = \underline{8.53 \text{ Kg/m}^2}$$

$$9.9) I = \frac{2}{5} M R^2 = \frac{2}{5} (28 \text{ Kg}) (0.38)^2 = 1.62 \text{ Kg/m}^2$$

$$K = \frac{1}{2} I \omega^2$$

$$\omega = \sqrt{\frac{2K}{I}}$$

$$\omega = \sqrt{\frac{2(170)}{1.62}} = 14.75 \text{ rad/s}$$

$$V = r \omega = 0.38 (14.75) = \underline{5.61 \text{ rad/s}}$$

$$9.7) M = 7.35 \times 10^{22} \text{ Kg}$$

$$R = 1.74 \times 10^6 \text{ m}$$

$$T_{rot} = 2\pi$$

$$27.3 \text{ d}$$

$$T_d = 8.64 \times 10^4 \text{ s}$$

$$a) I = \frac{2}{5} (7.35 \times 10^{22}) (1.74 \times 10^6)^2 = 8.9 \times 10^{34} \text{ Kg/m}^2$$

$$W = \frac{2\pi}{127.3 \times 8.64 \times 10^4} = 2.66 \times 10^{-4} \text{ rad/s} \quad K = \frac{1}{2} I \omega^2 = \frac{1}{2} (8.9 \times 10^{34}) (2.66 \times 10^{-4})^2 = \underline{3.15 \times 10^{13} \text{ J}}$$

$$b) \frac{3.15 \times 10^{23}}{5(4 \times 10^{20})} = 158 \text{ años}$$

$$9.43) \frac{1 \text{ rev}}{\text{min}} \times \frac{2\pi \text{ rad/s}}{60} \quad \Delta K = -500 \text{ J}$$

$$W_i = 650 \text{ rev/min} = 68.1 \text{ rad/s} \quad W_f = 520 \text{ rev/min} = 54.5 \text{ rad/s}$$

$$\Delta K = K_f - K_i = \frac{1}{2} I (W_f^2 - W_i^2)$$

$$I = \frac{2(\Delta K)}{W_f^2 - W_i^2} = \frac{2(-500 \text{ J})}{(54.5)^2 - (68.1)^2} = 0.60 \text{ Kg/m}^2$$

$$9.45) a_{\text{rad}} = R W^2$$

$$W = \sqrt{\frac{a}{R}} = \sqrt{\frac{3,500}{1.20}} = 54 \text{ rad/s}$$

$$\text{disco} = I = \frac{1}{2} M R^2 = \frac{1}{2} (70 \text{ Kg}) (1.20 \text{ m})^2 = 50.4 \text{ Kg/m}^2$$

$$K = \frac{1}{2} I W^2 = \frac{1}{2} (50.4) (54)^2 = 73,483.2 \text{ J}$$

$$9.47) a) K = \frac{1}{2} I W^2 \quad I = \frac{1}{2} M R^2$$

$$= \frac{1}{2} (2.5 \text{ Kg}) (0.200 \text{ m})^2 = 0.05 \text{ Kg/m}^2$$

$$W = \sqrt{\frac{2K}{I}}$$

$$= \sqrt{\frac{2(4.5 \text{ J})}{0.05 \text{ Kg/m}^2}} = 13.42 \text{ rad/s}$$

$$V = R W = (0.20 \text{ m}) (13.4 \text{ rad/s}) = 2.68 \text{ m/s}$$

$$K = \frac{1}{2} m v^2 = \frac{1}{2} (1.5 \text{ Kg}) (2.68 \text{ m/s})^2 = 5.39 \text{ J}$$

$$K_1 + U_1 = K_2 + U_2$$

$$0 = K_2 + U_2 \quad 0 = 4.5 + 5.39 + mg(-h)$$

$$h = \frac{9.89}{(11.5)(9.8)} = 0.67 \text{ m}$$

$$b) K_{\text{total}} = K + K = 9.89 \text{ J}$$

$$\frac{K}{K_{\text{total}}} = \frac{4.5}{9.89} = 0.45 \times 100 = 45\%$$

$$9.49) y - y_0 = \left(\frac{v_{y1} + v_{y2}}{2} \right) t$$

$$v = \frac{2(y - y_0)}{t} = \frac{2(3 \text{ m})}{2 \text{ s}} = 3 \text{ m/s}$$

$$\omega_{\text{zroc}} = \frac{v_y}{r} = \frac{3 \text{ m/s}}{0.280 \text{ m}} = 10.71 \text{ rad/s}$$

$$K_1 + U_1 + W = K_2 + U_2 \quad mg(3 \text{ m}) = \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2$$

$$I = \frac{2}{\omega^2} (mg(3 \text{ m}) - \frac{1}{2}mv^2) = \frac{2}{10.71^2} ((4.20)(9.8)(3) - \frac{1}{2}(4.20)(3)) = 7.82 \text{ kg}\cdot\text{m}^2$$

$$\text{disco} = I = \frac{1}{2}MR^2 \quad M = \frac{2I}{R^2} = \frac{2(7.82)}{0.280^2} = 46.5 \text{ kg}$$

$$9.51) a) K = \frac{1}{2}I\omega^2$$

la masa se multiplica = f^3
"distancia" " = f

$$f^3(f)^2 = f^5$$

$$b) (2.5 \text{ J})(48)^2 = 6.37 \times 10^8 \text{ J}$$

$$9.53) \quad I = I_{cm} + Md^2 \quad I = \frac{2}{3} MR^2$$

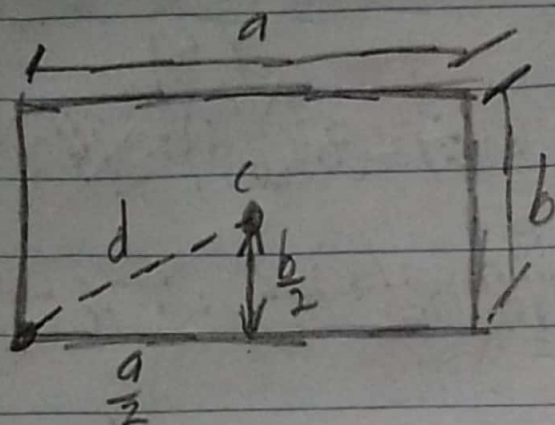
$$\frac{2}{3} MR^2 = \frac{2}{5} MR^2 + Md^2$$

$$\frac{2}{3} - \frac{2}{5} R^2 = d^2$$

$$d = \sqrt{\frac{(10-6)}{75}} = \frac{2R}{\sqrt{75}} = 0.52R$$

es paralelo a un
diámetro, 0.52R
del centro

9.55)



$$d = \sqrt{(a/2)^2 + (b/2)^2}$$

$$I = I_c + Md^2 = \frac{1}{12} M(a^2 + b^2) + M\left(\frac{1}{4}a^2 + \frac{1}{4}b^2\right) = \left(\frac{1}{12} + \frac{1}{4}\right) M(a^2 + b^2) = \frac{1}{3} M(a^2 + b^2)$$

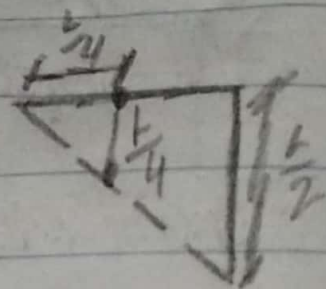
$$9.57) \quad a) \quad I_s = \frac{1}{3} \left(\frac{M}{2}\right) \left(\frac{L}{2}\right)^2 = \frac{1}{24} ML^2$$

$$I_{var} = 2I_s = \frac{1}{12} ML^2$$

$$b) \quad I = \frac{1}{12} \left(\frac{M}{2}\right) \left(\frac{L}{2}\right)^2 = \frac{1}{96} ML^2$$

$$I = \frac{1}{96} ML^2 + \frac{11}{2} \left(\frac{L}{4}\right)^2 = \frac{1}{24} ML^2$$

$$I = 2I_s = \frac{1}{12} ML^2$$



9.61) a) $\omega_z(t) = \omega_{0z} + \int_0^t \alpha_z(t') dt' = \int_0^t (8.6 \text{ rad/s}^2)t - (7.15 \text{ rad/s}^3)t^2 dt'$

At $t = 5 \text{ s}$ $\omega_z = (8.6)(5) - (7.15)(5) = \underline{74.2 \text{ rad/s}}$

b) $\theta - \theta_0 = \int_0^t \omega_z t' dt' = (4.3 \text{ rad/s}^2)t^2 - (0.38 \text{ rad/s}^3)t^3 \quad t = 5 \text{ s}$

$\theta - \theta_0 = 107.5 \text{ rad} - 47.9 \text{ rad} = \underline{59.6 \text{ rad}}$

9.63) $\theta - \theta_0 = (155 \text{ rev})(2\pi \text{ rad/rev}) = 973.9 \text{ rad}$
 $\alpha_z = (3 \text{ rev/s}^2)(2\pi \text{ rad/rev}) = 18.85 \text{ rad/s}^2$

$\omega_z^2 = \omega_{0z}^2 + 2\alpha_z(\theta - \theta_0)$

$\omega_z = \sqrt{2\alpha_z(\theta - \theta_0)} = \sqrt{2(18.85)(973.9)} = 191.6$

$v = r\omega = (0.120 \text{ m})(191.6 \text{ rad/s}) = 23 \text{ m/s}$

$y - y_0 = v_{0y}t + \frac{1}{2}at^2 \quad t = \sqrt{\frac{2(y - y_0)}{a_y}} = \sqrt{\frac{2(0.870)}{9.8}} = 0.41 \text{ s}$

$x - x_0 = v_{0x}t + \frac{1}{2}at^2 = (23 \text{ m/s})(0.41 \text{ s}) = \underline{9.43 \text{ m}}$

Ejercicios Cap. 10

10.1) a) $\tau = Fl$
 $l = r \sin \theta = (4\text{m}) \sin 90^\circ = 4\text{m}$

$$\tau = (10\text{N})(4\text{m}) = \underline{40\text{N/m}}$$

b) $\tau = Fl$
 $l = r \sin \theta = (4\text{m}) \sin 120^\circ = 3.46\text{m}$
 $\tau = (10\text{N})(3.46) = \underline{34.6\text{N/m}}$

c) $\tau = Fl$
 $l = r \sin \theta = (4) \sin 30^\circ$
 $l = 2\text{m}$
 $\tau = (10)(2) = \underline{20\text{N/m}}$

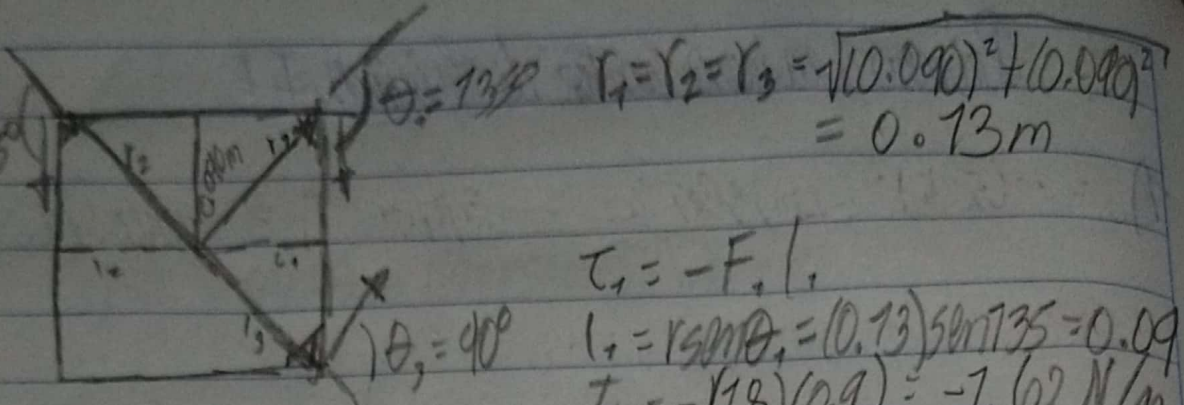
d) $\tau = Fl$
 $l = r \sin \theta = (2) \sin 60^\circ = 1.73\text{m}$
 $\tau = (10\text{N})(1.73) = \underline{17.3\text{N/m}}$

e) $\tau = Fl$
 $r = 0$
 $l = 0$
 $\tau = 0$

f) $\tau = Fl$
 $l = r \sin \theta =$
 $l = 0$
 $\theta = 180^\circ$
 $\tau = 0$

10.3)

$\theta = 135^\circ$



$\theta = 40^\circ$

$$\tau_1 = -F_1 l_1$$

$$l_1 = r \sin \theta = (0.13) \sin 135 = 0.09$$

$$\tau_1 = -(18)(0.09) = -1.62 \text{ N/m}$$

$$\tau_2 = F_2 l_2$$

$$l_2 = r \sin \theta = (0.13) \sin 135 = 0.09$$

$$\tau_2 = (26)(0.09) = 2.34 \text{ N/m}$$

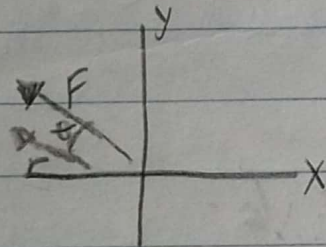
$$\tau_3 = F_3 l_3$$

$$l_3 = r \sin \theta = (0.13) \sin 40 = 0.13$$

$$\tau_3 = (14)(0.13) = 1.78 \text{ N/m}$$

$$\Sigma \tau = \tau_1 + \tau_2 + \tau_3 = -1.62 + 2.34 + 1.78 = 2.5 \text{ N/m}$$

10.5) a)



b)

$$\vec{c} = \vec{r} \times \vec{F} = (-0.45\hat{y}) + 0.15\hat{y}(-5\hat{j}) + 4\hat{j}$$

$$\vec{c} = (2.25)\hat{i} - (1.8)\hat{j} - 0.750\hat{j} + 0.600\hat{j}$$

$$\hat{i} \times \hat{j} = \hat{k} \quad \hat{j} \times \hat{j} = 0$$

$$\hat{i} \times \hat{j} = \hat{k}$$

$$\hat{j} \times \hat{j} = -\hat{k}$$

$$\vec{c} = -(1.8)\hat{k} - (0.750)(-\hat{k}) = (-1.05 \text{ N/m})\hat{k}$$

Norma

$$10.7) \tau_1 = (50\text{N})(0.2\text{m})\sin 60^\circ = \underline{8.7\text{N/m}}$$

$$\tau_c = -(50\text{N})(0.2\text{m})\sin 30^\circ = -5\text{N/m} \quad \tau_D = -(50)(0.2)\sin 90^\circ = -10\text{N/m}$$

$$b) \Sigma \tau = \tau_1 + \tau_B + \tau_c + \tau_D = -6.3\text{N/m}$$

$$10.1) \tau = I\alpha_z = I \frac{\omega_z - \omega_0}{t} = 2.5 \frac{47.9}{8} = \underline{13.1\text{N/m}}$$

$$10.11) \alpha_z = \frac{\tau_z}{I} = \frac{-(0.0200\text{N})(0.0150\text{m})}{\frac{2}{5}(0.225\text{kg})(0.0150\text{m})^2} = -14.8\text{rad/s}^2$$

$$b) \omega_z - \omega_0 = -22.5\text{rad/s}$$

$$\omega_z = \omega_0 + \alpha_z t \quad t = \frac{\omega_z - \omega_0}{\alpha_z} = \frac{-22.5}{-14.8} = \underline{1.52\text{s}}$$

$$10.13) x - x_0 = v_0 t + \frac{1}{2} a t^2$$

$$a = \frac{2(x - x_0)}{t^2} = \frac{2(1.2\text{m})}{(0.80\text{s})^2} = \underline{3.75\text{m/s}^2}$$

$$T_1 = m_1 a_1 = 7.5\text{N} \quad T_2 = m_2(g - a) = 18.2\text{N}$$

$$b) (T_2 - T_1)R = 0.80\text{N/m} \quad \alpha = \frac{a_1}{R} = 50\text{rad/s}^2$$

$$I = \frac{L}{\alpha} = \underline{0.016\text{kg/m}^2}$$

$$10.15) \omega_0 = 0 \quad \omega_z = 72 \text{ rev/s} = 75.40 \text{ rad/s}$$

$$\omega_z = \omega_0 + \alpha_z t \quad \alpha_z = \frac{\omega_z - \omega_0}{t} = \frac{75.40}{2} = 37.70 \text{ rad/s}^2$$

$$\Sigma \tau_z = I \alpha_z \quad I = \frac{\tau_z}{\alpha_z} = \frac{(80)(0.120)}{37.70} = 0.26 \text{ Kg/m}^2$$

$$10.17) 72 \text{ Kg} = \Sigma F_x = m a_x \quad T_1 = m_1 a$$

$$5 \text{ Kg} = \Sigma F_y = m a_y \quad m_2 g - T_2 = m_2 a$$

$$\Sigma \tau_z = I \alpha_z \quad (T_2 - T_1)R = \left(\frac{7}{2} M R^2\right) \alpha \quad a = R \alpha$$

$$T_2 - T_1 = \frac{7}{2} M a$$

$$m_2 g = (m_1 + m_2 + \frac{7}{2} M) a$$

$$a = \left(\frac{m_2}{m_1 + m_2 + \frac{7}{2} M} \right) g = \left(\frac{5}{72 + 5 + 7} \right) (9.8) = 2.72 \text{ m/s}^2$$

$$T_1 = (72)(2.72) = 32.6 \text{ N} \rightarrow \text{apoyando}$$

$$T_2 = m_2 (g - a) = (5)(9.8 - 2.72) = 35.4 \text{ N} \rightarrow \text{por debajo}$$

$$d) \underline{a = 2.72 \text{ m/s}^2}$$

$$c) \Sigma F_x = m a_x$$

$$\Sigma F_y = m a_y$$

$$F_x = T_1 = 32.6$$

$$F_y = Mg + T_2 = (72)(9.8) + 35.4 = 55 \text{ N}$$

20.19) a) $V_{cm} = RW = (0.600\text{m})(3\text{rad/s}) = \underline{1.8\text{m/s}}$

b) $K = \frac{1}{2}Mv^2 + \frac{1}{2}I\omega^2 = \frac{1}{2}Mv^2 + \frac{1}{2}(MR^2)\left(\frac{v}{R}\right)^2 = Mv^2 =$

$(2.2\text{kg})(1.8) = \underline{7.3\text{J}}$

c) i) $V = 2v = 3.6\text{m/s}$

ii) $V = 0$

iii) $V = \sqrt{v^2 + v^2} = \sqrt{v^2 + (RW)^2} = \sqrt{2}v = 2.55\text{m/s}$

d) i) $V = RW = 1.8\text{m/s}$ ii) $V = 1.8\text{m/s}$

iii) $V = 1.8\text{m/s}$

20.21) $\frac{(\frac{1}{2})I\omega^2}{(\frac{1}{2})Mv^2 + \frac{1}{2}I\omega^2} = \frac{1}{1 + (\frac{M}{I})v^2} = \frac{1}{1 + (MR^2/I)}$

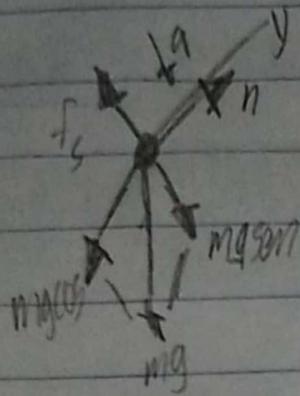
a) $I = \frac{1}{2}MR^2 = \underline{\frac{1}{3}}$

b) $I = \frac{2}{5}MR^2 = \underline{\frac{2}{7}}$

c) $I = \frac{2}{3}MR^2 = \underline{\frac{2}{5}}$

d) $I = \frac{5}{8}MR^2 = \underline{\frac{5}{13}}$

10.23) a)



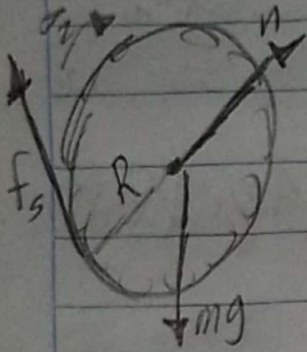
$$\sum F_y = ma_y$$

$$N = mg \cos \theta \quad f_s = \mu_s mg \cos \theta$$

$$\sum F_x = ma_x$$

$$mg \sin \theta - \mu_s mg \cos \theta = ma$$

$$g(\sin \theta - \mu_s \cos \theta) = a$$



$$\sum \tau = \tau_f = \mu_s mg \cos \theta R \quad I = \frac{2}{5} mR^2$$

$$\sum \tau = I\alpha \quad \mu_s mg \cos \theta R = \frac{2}{5} mR^2 \alpha$$

$$\alpha = \frac{a}{R}$$

$$\mu_s g \cos \theta = \frac{2}{5} a \quad a = \frac{5}{7} g \sin \theta$$

$$\mu_s = \frac{2}{7} \tan \theta = \frac{2}{7} \tan 15^\circ = 0.61$$

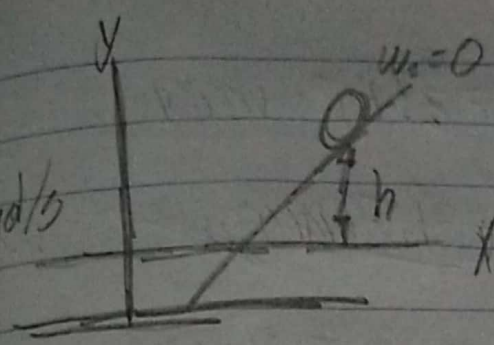
$$b) \mu_s = \frac{2}{5} \tan \theta$$

$$\mu_s = \frac{2}{5} \tan 15^\circ = 0.86$$

c) No hay deslizamiento donde hacen contacto, también se requiere más fricción para una bola hueca.

10.25)

$\omega = 2.5 \text{ rad/s}$



$$K_1 + U_1 + W = K_2 + U_2$$

$$W = W = -3500 \text{ J}$$

$$K_1 = \frac{1}{2} I \omega^2 + \frac{1}{2} M v^2$$

$$v = R \omega$$

$$I = 0.800 M R^2$$

$$K_1 = \frac{1}{2} (0.800) M R^2 \omega^2 + \frac{1}{2} M v^2 = 0.900 M R^2 \omega^2$$

$$K_2 = 0 \quad U_1 = 0 \quad U_2 = Mgh$$

$$0.900 M R^2 \omega^2 + W_f = Mgh$$

$$M = \frac{W}{g} = 3500 / 9.8 = 40 \text{ kg}$$

$$h = \frac{0.900 M R^2 \omega^2 + W_f}{Mg}$$

$$h = \frac{0.900 (40) (0.600)^2 (2.5)^2 - 3500 \text{ J}}{(40) (9.8)} = 11.7 \text{ m}$$

10.27)

a) $K_1 + U_1 = K_2 + U_2$

$$K_1 = 0$$

$$U_1 = 0$$

$$0 = U_2 + K_2$$

$$0 = -mgd + \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2 \quad \frac{1}{2} I \omega^2 = \frac{1}{2} \left(\frac{1}{2} m [R^2 + R^2] \right) v^2 / R^2$$

$$\frac{1}{4} m [1 + (R_0/R_0)^2] v^2$$

$$\frac{1}{2} \left(1 + \frac{1}{2} [1 + (R_0/R_0)^2] \right) v^2 = gd$$

$$d = \frac{\left(1 + \frac{1}{2} [1 + (R_0/R_0)^2] \right) v^2}{2g} = \frac{(140/40)(6/40)^2}{2(9.8)} = 3.7 \text{ m}$$

Norma

$$b) K_1 = \frac{1}{2}mv^2$$

$$mg = \frac{1}{2}mv^2$$

$$v = \sqrt{2gd} = \sqrt{2(9.8)(3.76)} = \underline{8.58 \text{ m/s}}$$

c) lo que para es que en el primer caso el cilindro tiene energía cinética, por lo mismo tiene menor velocidad.

$$10.29) a) K_1 + U_1 = K_2 + U_2 \quad U_1 = 0 \quad K_2 = 0$$

$$K_1 = U_2 \quad \frac{1}{2}mv^2 + \frac{1}{2}I\omega^2 = mgh$$

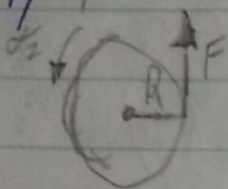
$$\frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{2}{3}mR^2\right)\left(\frac{v}{R}\right)^2 = \frac{1}{3}mv^2$$

$$\frac{5}{6}mv^2 = mgh \quad v = \sqrt{\frac{6gh}{5}} = \sqrt{\frac{6(9.8)(5)}{5}} = \underline{7.67 \text{ m/s}}$$

$$\omega = \frac{v}{R} = \frac{7.67}{0.11} = \underline{69.7 \text{ rad/s}}$$

$$b) K = \frac{1}{2}I\omega^2 = \frac{1}{3}mv^2 = \frac{1}{3}(0.426)(7.67)^2 = \underline{8.35 \text{ J}}$$

10.31) a)



$$\sum \tau_c = I\alpha_c$$

$$FR = I\alpha$$

$$\alpha = \frac{FR}{I} = \frac{(18)(2.4)}{2.100 \text{ kg}\cdot\text{m}^2} = 0.021 \text{ rad/s}^2$$

$$\omega_0 = 0 \quad t = 15 \text{ s}$$

$$\omega_2 = \omega_0 + \alpha t = 0 + (0.021)(15) = \underline{0.31 \text{ rad/s}}$$

$$b) W_1 = \tau_c \Delta\theta$$

$$\Delta\theta = \theta - \theta_0 = \omega_0 t + \frac{1}{2}\alpha t^2 = 0 + \frac{1}{2}(0.021)(15)^2 = \underline{2.31 \text{ rad}}$$

$$\tau_z = FR = (18)(2.4) = 43.2 \text{ N/m}$$

$$W = \tau_z \Delta\theta = (43.2)(2.31) = 100 \text{ J}$$

$$c) \text{Potencia} = \frac{\Delta W}{\Delta t} = \frac{100 \text{ J}}{15.5} = 6.47 \text{ W}$$

$$10.33) a) \tau_z = I \alpha_z = I \frac{\omega_z - \omega_0}{t} =$$

$$\tau_z = \left(\frac{(1/2)(1.5)(0.100)^2}{2.5} \right) (1200) \left(\frac{\pi}{30} \right) = 0.38 \text{ m}$$

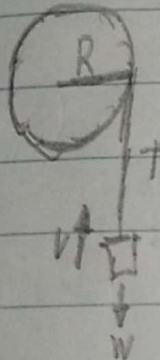
$$b) \omega \Delta t = \frac{(600)(2.5)}{60} = 25 \text{ rev} = 157 \text{ rad}$$

$$c) W = \tau \Delta\theta = (0.38)(157) = 59.2 \text{ J}$$

$$d) K = \frac{1}{2} I \omega^2 = \frac{1}{2} \left((1/2)(1.5)(0.100)^2 \right) \left((1200) \left(\frac{\pi}{30} \right) \right)^2 = 59.2 \text{ J}$$

$$10.35) a) \omega_z = (4000) \left(2\pi / 1 \text{ rev} \right) (1 \text{ min} / 60 \text{ s}) = 418.9 \text{ rad/s}$$

$$\tau_z = \frac{P}{\omega_z} = \frac{1.5 \times 10^5 \text{ W}}{418.9} = 358 \text{ N/m}$$

b)  $T = W$ $\tau_z = TR$
 $T = \frac{\tau_z}{R} = 358 / 0.100 = 3580 \text{ N}$ $W = 3580 \text{ N}$

$$c) v = R\omega$$

$$v = (0.100)(418.9) = 41.9 \text{ m/s}$$

$$10.37) L = mvr \sin \theta = (2)(12)(8) \sin 143.1^\circ = 175 \text{ Kg/m}^2/\text{s}$$

$$b) \tau = mg(8\text{m}) \cos 36.9^\circ = 125 \text{ Kg} \cdot \text{m}^2/\text{s}^2$$

$$10.39) L = I \omega$$

$$\omega = 1 \frac{\text{rev}}{\text{min}} \cdot \frac{2\pi \text{ rad}}{1 \text{ rev}} \cdot \frac{1 \text{ min}}{60 \text{ s}} = 0.105 \text{ rad/s}$$

$$I = \frac{1}{3} ML^2 = \frac{1}{3} (6 \times 10^{-3}) (0.150)^2 = 4.5 \times 10^{-5} \text{ Kg/m}^2$$

$$L = (4.5 \times 10^{-5}) (0.11) = 4.95 \times 10^{-6} \text{ Kg/m}^2/\text{s}$$

$$10.41) \omega_2 = \omega_1 \left(\frac{R_1}{R_2} \right)^2 = \left(\frac{2\pi}{(300)(86,400)} \right) \left(\frac{7 \times 10^5}{10} \right)^2 = 4.6 \times 10^3 \text{ rad/s}$$

$$10.43) L_i = L_f$$

$$I_i \omega_i = I_f \omega_f$$

$$I_i = 0.4 + \frac{1}{12} (8)(7.8)^2 = 2.56 \text{ Kg/m}^2$$

$$I_f = 0.4 + 8(0.25)^2 = 0.90 \text{ Kg/m}^2$$

$$\omega_f = \left(\frac{I_i}{I_f} \right) \omega_i = \left(\frac{2.56}{0.9} \right) (0.4) = 1.14 \text{ rev/s}$$

$$10.45) a) L_1 = L_2$$

$$I_1 \omega_1 = I_2 \omega_2$$

$$\omega_2 = \left(\frac{I_1}{I_2} \right) \omega_1$$

$$I_1 = I_2 = \frac{1}{2} MR^2 = \frac{1}{2} (120)(2)^2 = 240 \text{ Kg/m}^2$$

$$I_2 = I_1 + I_p = 240 + mR^2 = 240 + (70)(2)^2 = 520 \text{ Kg/m}^2$$

$$\omega_2 = (240/520) 3 = 1.38 \text{ rad/s}$$

$$b) K_1 = \frac{1}{2} I_1 \omega_1^2 = \frac{1}{2} (240)(3)^2 = 1080 \text{ J}$$

$$K_2 = \frac{1}{2} I_2 \omega_2^2 = \frac{1}{2} (520)(1.38)^2 = 495 \text{ J}$$

Norma

10.47) a) $L_2 = mvr - I\omega$

$I = \frac{1}{3} M R^2$

$L_1 = L_2 \quad mvr = \frac{1}{3} m r^2 \omega$

$\omega = \frac{3mv}{mr} = 0.12 \text{ rad/s}$

b) $K_1 = 0$

$K_2 = \frac{1}{2} m v^2 + \frac{1}{2} I \omega^2$

$K_2 = \frac{1}{2} (10.0100) (0.200)^2 + \frac{1}{2} \left[\frac{1}{3} (0.0500) (1)^2 \right] (0.120)^2 = 3.2 \times 10^{-4} \text{ J}$

c) El aumento de la energía proviene del trabajo que realiza el hilo al empujarlo

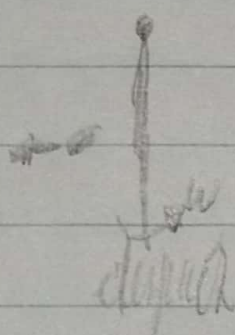
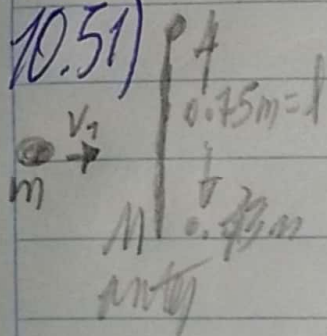
10.49) a)

$m_1 v_1 d = -m_2 v_2 d + \frac{1}{2} m L^2 \omega$
 $(3)(10)(7.5) = -(3)(6)(7.5) + \frac{1}{2} \left(\frac{90}{4.8} \right) (2)^2 \omega$

$\omega = 5.88 \text{ rad/s}$

b) El momento angular se conserva porque no hay torques desequilibrados y el momento lineal no se conserva porque el pivote ejerce una fuerza externa.

10.51)



a) $L_1 = L_2 \quad mvl = -m v_2 l + L \omega$

$\omega = \frac{3m(v_1 + v_2)}{2Ml} = \frac{3(1.1)(5+2)}{2(45)(1.5)} = 1.71 \text{ rad/s}$

b) No hay fuerza externa por lo que el momento lineal no se conserva, como tampoco hay par externo el momento angular no se conserva.

Norma

10.53) a) $F_p = W = (0.165)(9.8) = \underline{1.62 \text{ N}}$

b) $\omega = \frac{wr}{I\Omega} = \frac{(0.165)(9.8)(0.04)}{(7.2 \times 10^{-4})(2.857)} = 789 \text{ rad/s} = \underline{7.8 \times 10^3 \text{ rev/min}}$

10.55) a) mitad

b) si r y I no cambian, duplicado

c) si w y r no se cambian, se reduce a la mitad

d) doble

e) se queda igual